



Python File I/O

Data with Dylan | Cheat Sheet | Python Fundamentals #6

WHAT IS FILE I/O?

File I/O moves data between Python and files on your computer.

💡 Like a notebook: open it, read or add notes, then close it.

OPEN + CLOSE FILES

Traditional

```
file = open("notes.txt", "w")
file.write("Hello\n")
file.close()
```

Preferred

```
with open("notes.txt", "w") as file:
    file.write("Hello\n")
```

💡 ``with`` closes the file automatically, even after an error.

READ TEXT FILES

```
with open("notes.txt", "r") as file:
    text = file.read()          # entire file
    # lines = file.readlines()  # list of lines
```

TYPE / FORMATTING WARNINGS

- ▲ No ``close()`` manually -> data may not finish saving.
- ▲ ``r`` + missing filename/path -> `FileNotFoundError`.
- ▲ ``w`` clears old data. ``a`` keeps it and adds at the end.
- ▲ CSV: add ``newline=""`` to prevent extra blank rows.

⚡ MINI PRACTICE

Create ``college_log.txt``.

1) Append: UW | 13406 | 42.5

2) Reopen the file and print every entry.

Hint: `"a" + "\n" -> "r" + .read()`

FILE ACCESS MODES

MODE	WHAT IT DOES	EXISTS / MISSING
r	Read contents	Reads / <code>FileNotFoundError</code>
w	Write new contents	Clears / Creates file
a	Add at the end	Keeps old data / Creates file

KEY SYNTAX & OPERATIONS

Write vs. append

```
with open("log.txt", "w") as f:
    f.write("First entry\n") # clears old
with open("log.txt", "a") as f:
    f.write("Next entry\n") # adds at end
```

DICTIONARY -> STRUCTURED CSV

CSV = comma-separated values. Keys become column headers.

One row dictionary holds one college record.

```
import csv
with open("college_data.csv", "w", newline="") as f:
    writer = csv.DictWriter(f, fieldnames=fields)
    writer.writeheader()
```

Write each row

```
for i in range(num_rows):
    row_dict = {key: college_data[key][i] for key in fields}
    writer.writerow(row_dict)
```

💡 ``writerow()`` matches row dictionary keys to the fields.

COMMON BEGINNER MISTAKES

WHAT GOES WRONG	SAFER APPROACH
✗ Forgot to close the file file = open("log.txt", "a") file.write("UW\n") # Must call file.close() yourself.	✓ <code>`with`</code> closes it for you with open("log.txt", "a") as f: f.write("UW\n") # Closes after this block.
✗ "w" deletes old contents with open("log.txt", "w") as f: f.write("New entry\n") # Before: UW, WSU -> New entry	✓ "a" keeps old contents with open("log.txt", "a") as f: f.write("New entry\n") # After: UW, WSU, New entry
✗ Tried to read a missing file open("missing.txt", "r") # FileNotFoundError	✓ Confirm the file exists first with open("log.txt", "r") as f: print(f.read()) # Check spelling, folder, and .txt
Memory rule: with closes • r reads existing • w wipes + writes • a adds	